# Pritam Das

ee25mtech11021@iith.ac.in | pritamd61@gmail.com | +91 8017142759 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 6.67 |
| JLD College Of Engineering And Management | Class XII (West Bengal State Council Of Technical & Vocational Education And Skill Development) | 2017 | 7.70% |
| Nimta High School | Class X (West Bengal BVoard Of Secondary vEducation) | 2012 | 61.00% |

## WORK EXPERIENCE

**Teaching Assistant - Control Systems | Indian Institute of Technology Hyderabad (IITH)**

- Assisted in teaching, tutorials, laboratory sessions, and student support for the Control Systems course. -
- Guided students in control-system concepts, problem-solving, and academic activities.

**Trainee Engineer | Star Track Automation**

- ***Industrial Automation:*** Developed and implemented PLC programs, HMI/SCADA applications, and industrial control systems using Rockwell and Siemens automation platforms.
- ***Process Control & Integration:*** Integrated sensors, actuators, and industrial communication systems to support process automation, system monitoring, and operational efficiency.

**Safety Engineer(Driver Coach) | Air Water India Pvt Ltd**

- Vehicle Safety & Compliance: Conducted safety inspections of cryogenic product transport vehicles and prepared safety parameter reports to ensure operational compliance.
- Operations & Safety Management: Managed CESG store operations, ECR/JDE distribution invoicing, driver safety training, and RFID-based cylinder tracking and vehicle CCTV monitoring systems.

**Purchase | Bkg Electro Fab**

- ***Procurement & Vendor Management:*** Managed procurement activities for the cryogenic industry, coordinating with vendors and clients to ensure timely sourcing, cost-effective purchasing, and smooth supply operations.

## POSITIONS OF RESPONSIBILITY

**Placement Coordinator - Systems and Control, Electrical Engineering**

- Coordinate placement-related communication, recruiter interactions, and placement activities for the cohort, ensuring smooth coordination between students and the placement team.

**President - Bengali Community, IIT Hyderabad**

- Lead cultural events and community initiatives for the Bengali student community, coordinating teams, programs, and student engagement activities.

**Multimedia Coordinator - EML Club**

- Manage multimedia content, event coverage, and promotional materials for EML Club events.

**Campus Tour Head - Prayas | Rural Development Cell (RDC), IIT Hyderabad** |

- Lead and oversee campus tours, rural outreach, educational initiatives, student engagement, and visitor coordination for prospective students and rural communities.

**Coordinator - Prayas | Rural Development Cell (RDC), IIT Hyderabad** |

- Coordinated and executed rural outreach programs, educational campus visits, event logistics, stakeholder engagement, and on-ground team operations.

## SKILLS

**Project & Simulation Software** :MATLAB, Simulink, Python, Arduino IDE, LTSPICE, Cadence, Xilinx Vivado, Verilog, ,

OpenCV.

**Programming & Hardware Description Languages** : Python, C, Core Java, Verilog.

**Industrial & Job Experience Skills** :PLC Programming (Rockwell, Siemens), HMI/SCADA, AutoCAD, Safety Compliance & Reporting, Purchase & Procurement (Tally, JDE), Vendor Coordination, RFID & CCTV Systems, Java, Automation.

**Non-Tech & Job Skills :**Team Collaboration, Leadership, Public Speaking, Teaching & Mentoring, Event Management, Content Creation, Problem-Solving, Time Management, Communication, Adaptability.

**language:** Bengali, Hindi, English.

## RELEVANT COURSES

**Systems & Control**: *State Space, State Estimation, Optimal Control, Advanced Control Lab*

**Mathematics & Optimization**: *Matrix Theory, Random Variables & Stochastic Processes, Convex Optimization*

**AI/ML**: *Machine Learning for Process Systems Engineering, Pattern Recognition & Machine Learning, Machine Learning & Its Applications*

**Power & Energy**: *Smart Grid, Energy Management*

**VLSI & Digital Systems**: *Digital IC Design Lab, Memory Circuits & System Design*

**Automotive Systems**: *Automotive Communication & Sensing*

## PROJECTS

**Master's Thesis Project**

***Evaluating Criticality of Nodes in Consensus Networks Under False Data Injection Attacks***

***Thesis | May 2026 - Present***

***Present Work:***

- Analyze **finite-duration, magnitude-bounded FDI attacks** and their impact on multi-agent consensus.
- Derive **worst-case attack inputs** and **Induced Terminal Disagreement (ITD)** for node-criticality assessment.

**Future Work:**

- Develop **resilient consensus and FDI attack-mitigation strategies**.
- Investigate **distributed attack detection and mitigation** for multi-agent systems.

**Tools:** MATLAB, Python

**ACADEMIC PROJECTS**

***Reading External Analog Signals Using Xilinx XADC on Zynq-7000 FPGAs***

*Digital IC Designing Lab | Course Project | Jan-Apr 2026*

- Configured **XADC Wizard IP** for external analog signal acquisition and conversion.
- Designed and verified the acquisition interface using **Icarus Verilog**.

- **Tools:** Vivado, Verilog, Icarus Verilog

***Safety-Critical Control via Convex Control Barrier Function Synthesis***

*Convex Optimization | Course Project | Jan-Apr 2026*

- Studied **CBF-based safety constraints** using convex optimization for dynamical systems.
- Analyzed integration with **QP-based safety-critical controllers**.
- **Tools:** MATLAB, Python

***Partial Shading Mitigation in Photovoltaic (PV) Systems***

*Smart Grid | Course Project | Jan-Apr 2026*

- Analyzed partial shading effects on **PV power output and efficiency**.
- Studied techniques for **power-loss reduction and performance improvement**.
- **Tools:** MATLAB, Simulink

***Multi-View Perception for Autonomous Vehicles Using 360° Vision & Satellite Imagery***

*Automotive Communication & Sensing | Course Project | Jan-Apr 2026*

- Developed a **360° surround-view image analysis** framework for autonomous vehicles.
- Performed **BEV generation, image stitching, and image-quality analysis** using satellite/road imagery.
- **Tools:** Python, OpenCV, Google Colab

***IoT-Based Home Automation***

*B.Tech Project | 2019-2020*

- Designed an **Arduino Uno-based** home automation system for appliance control.
- Implemented **sensor-based monitoring and remote control**.
- **Tools:** Arduino Uno, Arduino IDE

## EXTRACURRICULAR

**Interests** - Robotics-based projects

**Activities** - Actively organized and coordinated technical fests and events during college, including:

- **Event Coordinator, MSME Tech Connect** -- Coordinated event planning, stakeholder communication, logistics, and on-ground execution.
- **Event Coordinator, NCC Event** -- Managed event coordination, logistics, and smooth execution in collaboration with the organizing team.
- **Organising Team Member, EML Event featuring Shubhanshu Shukla** -- Supported event planning, coordination, and on-ground execution.

## AWARDS & CERTIFICATE

Certificate of Appreciation - Recognized as the Best (Most Responsible & Disciplined) Student of the Electrical Engineering Department, RCC Institute Of Information Technology, Batch 2016-2020.

## HOBBIES & INTERESTS

- **Robotics & Automation**: Exploring robotics and industrial automation technologies.
- **Control Systems:** Learning about controller design and practical applications.
- **Content Creation:** Creating engaging digital content.
- **Travel & Exploration**: Discovering new places and cultures.
- **Event Management:** Organizing technical and cultural events.
- **Sports:** Playing cricket and badminton, and enjoying chess and carrom.
- **Social Volunteering**: Mentoring rural students on higher education and GATE preparation..
- **Cooking**: Preparing traditional dishes,